

2024.

Q1.

$R_0 \rightarrow A$
 $e \rightarrow B$

$L_0 \rightarrow C$
 $\lambda_0 \rightarrow D$

$L_B = E$
 $\lambda_B = F$

(a). Glasgow: $L_G = 55^\circ 52' 14'' = 55.871944^\circ$

$$\lambda_G = -4^\circ 25' 59'' = -4.433056^\circ$$

Beijing: $L_B = 40^\circ 4' 21'' = 40.072500^\circ$

$$\lambda_B = 116^\circ 35' 51'' = 116.597500^\circ$$

The radius of curvature at each end:

$$R_E(L_G) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_G}} = 6,392,816.37 \text{ m}$$

$$R_E(L_B) = \frac{R_0}{\sqrt{1 - e^2 \sin^2 L_B}} = 6,387,002.89 \text{ m}$$

For Glasgow:

$$x_{EB}^E = R_E(L_G) \cos L_G \cos \lambda_G = 3,575,923.71 \text{ m}$$

$$y_{EB}^E = R_E(L_G) \cos L_G \sin \lambda_G = -277,227.72 \text{ m}$$

$$z_{EB}^E = (1 - e^2) R_E(L_G) \sin L_G = 5,256,456.15 \text{ m}$$

For Beijing:

$$x_{EB}^E = R_E(L_B) \cos L_B \cos \lambda_B = -2,188,244.44 \text{ m}$$

$$y_{EB}^E = R_E(L_B) \cos L_B \sin \lambda_B = 4,370,300.45 \text{ m}$$

$$z_{EB}^E = (1 - e^2) R_E(L_B) \sin L_B = \underline{4,084,149.01 \text{ m}}$$

According to: $\underline{a} \cdot \underline{b} = |\underline{a}| \cdot |\underline{b}| \cdot \cos \theta$

$$\Rightarrow \cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| \cdot |\underline{b}|}$$

The angle between Glasgow and Beijing is

$$\Rightarrow \theta = 72.138584^\circ$$

Therefore, the distance is

$$\text{distance} = 40030.2 \text{ km} \cdot \frac{\theta}{360} = 8021.45 \text{ km}$$

(b). The absolute acc. we seek is

$$\underline{a}_{ip}^b = \underline{C}_i^b \cdot \ddot{\underline{r}}_{ip}^i$$

$$\underline{r}_{ip}^i = \underline{r}_{ib}^i + \underline{r}_{bp}^i = \underline{r}_{ib}^i + \underline{C}_b^i \underline{r}_{bp}^b$$

$$\Rightarrow \ddot{\underline{r}}_{ip}^i = \ddot{\underline{r}}_{ib}^i + \ddot{\underline{C}}_b^i \underline{r}_{bp}^b + \underline{C}_b^i \ddot{\underline{r}}_{bp}^b \quad (\underline{r}_{bp}^b = \underline{r}_{bp}^b)$$

$$= \underline{C}_b^i \underline{u}_b - \underline{\Omega}_{bi}^i \underline{C}_b^i \underline{r}_{bp}^b + 0 \quad (\underline{u}_b = \underline{C}_i^b \ddot{\underline{r}}_{ib}^i \Rightarrow \ddot{\underline{r}}_{ib}^i = \underline{C}_b^i \underline{u}_b)$$

$$\Rightarrow \ddot{\underline{r}}_{ip}^i = \underline{C}_b^i \underline{\Omega}_{ib}^b \underline{u}_b + \underline{C}_b^i \underline{u}_b - \underline{\Omega}_{bi}^i \underline{C}_b^i \underline{r}_{bp}^b + \underline{\Omega}_{bi}^i \underline{\Omega}_{bi}^i \underline{C}_b^i \underline{r}_{bp}^b + 0$$

$$\cancel{\underline{\Omega}_{bi}^i \underline{C}_b^i = \underline{C}_b^i \underline{\Omega}_{bi}^b}$$

$$\Rightarrow \underline{a}_{ip}^b = \underline{C}_i^b \ddot{\underline{r}}_{ip}^i = \underline{\Omega}_{ib}^b \underline{u}_b + \underline{u}_b - \underline{\Omega}_{bi}^b \underline{r}_{bp}^b + \underline{\Omega}_{bi}^b \underline{\Omega}_{bi}^b \underline{r}_{bp}^b$$

Q2.

(a) i. Strapdown Equation:

$$\dot{C}_b^n = C_b^n \Omega_{nb}^b \quad \Omega_{nb}^b = \Omega_{ib}^b - (\Omega_{ie}^b + \Omega_{en}^b)$$

$$\begin{aligned} \underline{\omega}_{nb}^b &= \underline{\omega}_{ib}^b - \underline{\omega}_{in}^b \\ &= \underline{\omega}_{ib}^b - \underline{\omega}_{ie}^b - \underline{\omega}_{en}^b \end{aligned}$$

$$\Rightarrow \Omega_{nb}^b = \Omega_{ib}^b - (\Omega_{ie}^b + \Omega_{en}^b)$$

$$\Rightarrow \dot{C}_b^n = C_b^n \Omega_{ib}^b - C_b^n \Omega_{ie}^b - C_b^n \Omega_{en}^b$$

$$\text{Since } C_b^n \Omega_{ie}^b C_n^b = \Omega_{ie}^n \Rightarrow C_b^n \Omega_{ie}^b = \Omega_{ie}^n C_b^n$$

$$C_b^n \Omega_{en}^b C_n^b = \Omega_{en}^n \Rightarrow C_b^n \Omega_{en}^b = \Omega_{en}^n C_b^n$$

$$\text{Therefore } \dot{C}_b^n = C_b^n \Omega_{ib}^b - (\Omega_{ie}^n + \Omega_{en}^n) C_b^n$$

ii. Navigation Equation:

$$\dot{C}_{ib}^n = \dot{f}_{ib}^n + \underline{y}_{ib}^n$$

what we want to get is $\dot{V}_{eb}^n$

$$\underline{V}_{eb}^n = C_e^n \underline{V}_{eb}^e$$

$$\Rightarrow \dot{\underline{V}}_{eb}^n = \dot{C}_e^n \underline{V}_{eb}^e + C_e^n \dot{\underline{V}}_{eb}^e$$

$$= -\Omega_{en}^n C_e^n \underline{V}_{eb}^e + C_e^n (-\Omega_{ie}^e \Omega_{ie}^e \underline{r}_{eb}^e - 2\Omega_{ie}^e \underline{V}_{eb}^e + \underline{a}_{ib}^e)$$

$$= -\Omega_{en}^n \underline{V}_{eb}^n - \Omega_{ie}^n \Omega_{ie}^n \underline{r}_{eb}^n - 2\Omega_{ie}^n \underline{V}_{eb}^n + \underline{a}_{ib}^n$$

$$= -\Omega_{ie}^n \Omega_{ie}^n \underline{r}_{eb}^n - (\Omega_{en}^n + 2\Omega_{ie}^n) \underline{V}_{eb}^n + \underline{a}_{ib}^n$$

$$\Rightarrow \dot{\underline{V}}_{eb}^n = C_b^n \underline{f}_{ib}^b + C_b^n \underline{y}_{ib}^b - \Omega \Omega \underline{r} - (\Omega + 2\Omega) \underline{V}$$

$$= C_b^n \underline{f}_{ib}^b + \underline{y}_b^n (L_b, h_b) - (\Omega_{en}^n + 2\Omega_{ie}^n) \underline{V}_{eb}^n$$

(b) 4 Error:

(1) Bias : constant offset in the output.

(2) Scale factor error : gradient of the curve relating I/O

(3) Cross-coupling : misalignment of sensors sensitive axis with orthogonal body axis

(4) Random noise : Thermo-mechanical White noise

$$\begin{aligned} \text{(c)} \quad \overset{\text{measured}}{\tilde{x}} &= b + (I + M)x + W \\ \Rightarrow \begin{cases} \hat{f} = b_a + (I + m_a)f + w_a \\ \hat{w} = b_g + (I + m_g)w + w_g \end{cases} \end{aligned}$$

Q3. (a).

True range:

$$B_{1T} = \sqrt{(60-20)^2 + (70-25)^2} = 60.207973$$

$$B_{2T} = \sqrt{(-50-20)^2 + (-45-25)^2} = 98.994949$$

$$B_{3T} = \sqrt{(150-20)^2 + (-80-25)^2} = 167.167750$$

Iteration 1: (position estimate $\underline{d_0} = (0, 0)$)

$$R_1 = \|\underline{B}_1 - \underline{d}_0\| = 92.195445$$

$$R_2 = \|\underline{B}_2 - \underline{d}_0\| = 67.268120$$

$$R_3 = \|\underline{B}_3 - \underline{d}_0\| = 170.000000$$

its Jacobian matrix:

$$H = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix} \approx \begin{bmatrix} \frac{0-60}{92.195445} & \frac{0-70}{92.195445} \end{bmatrix}$$

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{0+50}{67.268120} & \frac{0+45}{67.268120} \\ \frac{0-150}{170.000000} & \frac{0+20}{170.000000} \end{pmatrix}$$

$$= \begin{bmatrix} -0.650791 & -0.759257 \\ 0.743294 & 0.668965 \\ -0.882253 & 0.470588 \end{bmatrix}$$

The error:

$$\delta R = \begin{bmatrix} 60.207473 - 92.195445 \\ 98.994444 - 67.268120 \\ 167.107750 - 170.000000 \end{bmatrix}$$

$$= \begin{bmatrix} -31.987472 \\ 31.726824 \\ -2.892250 \end{bmatrix}$$

$$\Rightarrow \delta u = (H^T H)^{-1} H^T \delta R = \begin{bmatrix} 17.822196 \\ 27.207475 \end{bmatrix}$$

$$\Rightarrow u_1 = u_0 + \delta u = (17.822196, 27.207475)$$

$$\|g_n\| = 32,525027 > 0.1$$

Iteration 2 :